

sector, and we seem to be catching up with the cutting-edge technology of the world with great difficulty. The pace and spread of current technological developments have overshadowed our history and make us believe that the ancestral societies other than the West have very little to contribute to the S & T ecosystem. We also tend to believe that all that we have in the last 250–500 years are the only living example of S & T. On account of these, an average well-educated Indian believes that Science and Technology are of Western origin and India has very little to contribute. If we go by this logic, then the medieval society ought to have had a very simple and primitive living in which S & T played practically no role. In this chapter and a few that follow, we focus on this aspect.

10.1 THE INDIAN S & T HERITAGE

Indian civilisation has a rich heritage spanning over the recorded history of the last two millennia. Several archaeological and literary resources point to the existence of a vibrant community in antiquity. Several artifacts, relics, and physical structures dot the length and breadth of the country that bears testimony to this. In the earlier chapters of the book, we discussed in some detail the scientific heritage of ancient Indians and their contributions. We were able to see how with a well-developed body of knowledge in Mathematics, Astronomy, and other fields several theoretical foundations were laid for the benefit of the scientific community. However, we are not aware if Engineering and Technology were part of ancient Indian society. Did our ancestors have any knowledge of S & T? If so, how did they employ the knowledge and how did they benefit from it?

We can approach this question by exploring if there is any evidence of the use of S & T in ancient times. The evidence could be from any of the following three:

- (a) The archaeological evidence that shows how S & T ought to have been deployed in ancient times
- (b) Living examples of physical entities that point to the use of good S & T practices
- (c) Literary resources pointing to the use of S & T in ancient times

In this section of the book, we deploy all three of these approaches to understand the S & T heritage of ancient Indians. We explore a variety of available evidence to make sense of the S & T heritage that could have existed in ancient and medieval times in India. Furthermore, we explore the available knowledge sources in the domain of S & T. Using these we discuss various aspects of S & T in Chapters 11 and 12. A related field of Town Planning and Architecture is separately discussed in Chapter 12.

Archaeological excavations in the last 150 years in India have fetched a rich collection of artefacts that reveal several clues to the Engineering and Technology heritage of ancient Indians. Amongst other things, the excavations of Kalibangan in Rajasthan and Lothal in Gujarat have revealed functional town-planning, dwelling houses built of different sizes of burnt bricks but of a standard proportion of dimensions, tiled flooring, well-developed drainage systems, and pottery which are both utilitarian as well as decorated. Furthermore, it also points to agricultural operations, shipbuilding, metalworking, especially iron and steel, copper, and zinc, and a host of other skilled crafts such as intricate bead-making, and jewellery among the people. Excavations show that Lothal had a man-made dock for berthing boats (or smaller size ships)—a trapezoid basin measuring about 214 m × 36 m. What appears to be the largest maritime structure of the times, had facilities for loading and unloading¹. There are studies on town planning and architectural developments, skill, and developments.

Archaeological and literary evidence suggests that using the technology they were making military equipment, agricultural implements, ornamental pieces (see Figure 10.1 for an example from 1st century BCE), idols, coins, and a host of other items for medical applications. They were also adept at extracting Mercury and using it in alchemy. Several travellers' accounts (such as that of Herodotus, Ktesias, Hiuen Tsang, and Pliny) reveal how famous the Indian metalsmiths were for their quality and workmanship. We shall see some details of these aspects in Chapter 11.

10.2 MINING AND ORE EXTRACTION

In technology, ancient Indian's contributions to metalworking, metallurgy, and material science are noteworthy. Indian metalsmiths had notable achievements, particularly developing unique skills in mining and working with iron and steel, copper and its alloys, and zinc. Archaeological examination of the vicinity of the major mining centres of modern India viz., Zawar, Rajpura-Dariba, Rampura-Agucha, Khetri, Chamba, Deri-Ambaji, Singhbhum, Chitradurga, and Cudappah presents an interesting picture. These locations have extensive ancient mine workings and debris, and heaps of slags and retorts. Furthermore, a study of the ruins of temples and townships in various places in India corroborates the degree to which mining, and metalworking have developed in ancient India. From an analysis of the archaeological specimens obtained one can infer the skill and technical superiority exhibited by our ancestors in metallurgical works, which we seem to be largely unaware of.



FIGURE 10.1 Andhra Pradesh Royal Earrings (1st Century BCE)
Source: PHGCOM, Photographed at the Metropolitan Museum of Art.

a few steps. First, the ore was dislodged by setting the fire. Further, it was quenched in water, and using a chisel and hammer it was extracted. These extracted ores were carried out of the mines by the labourers in their heads for further processing.

To crack the ore, large quantities (about 6–7 tonnes) are stacked and set fire. The workers immediately leave the mines only to return after three days. The studies on the mining process suggest that each worker proceeded into the mine with a chisel, lamp, and a small basket. The lamp placed on his head provided not only illumination but also helped him identify the glittering particles of ore. While at work the miner seated upon his heels with the lamp upon his head, the hammer in his right hand and the chisel in his left hand, and the small basket upon his knee in which he received all the fragments of ore that were struck off by the chisel. The basket was passed through a chain of workers, and in this manner, the ore was cascaded to the surface eventually⁵.

- In the Vedic texts, we find ample references to Gold, Silver, iron, copper, and their alloys.
- The Indian metal smiths were adept in alloy technology as they could produce alloys of controlled composition.

Once the ore is brought to the surface, they were finely powdered using heavy hammers, so that they are conducive for the roasting process. The powdered ore was mixed with cow-dung and made into rolls about four inches long which were dried first in the Sun and then roasted in the open air in a fire of cow-dung cake. The ore was then ready for smelting. There is enough evidence to show the smelting of the ore using locally built furnaces, cemented with clay having nozzles and bellows. Using a pair of sticks, the bellow valve was built and they were opened when the bag was raised for blowing the air. Through a hole made at the bottom of the furnace, the molten metal was stirred and drawn out. After lighting the furnace, the ore was gradually introduced alternatively with charcoal and flux (refuse from the old iron furnaces).

10.3 METALS AND METALWORKING TECHNOLOGY

In modern times metals and alloys play an important role in our our lives. Several engineering gadgets and equipment make use of a variety of metals and their alloys. Ancient Indians have made use of base metals and alloys in multiple ways. This includes the making of military aids, ornaments, vessels, tools, artefacts, statues, and coins to name a few. In particular, Indians were known to be quite advanced in iron and steel, and zinc. The archaeological findings of Harappa and Mohenjo-Daro as well as in burial sites in South India have artefacts in Gold, Silver, Copper, and Bronze. The bronze statue of a dancing girl found at Mohenjo-Daro bears testimony to the technological skills of the ancient craftsmen. The statue is evidence of the skills of the craftsmen in drilling fine holes and casting the statue using the lost wax process (known in modern times as the cire-perdue process). In the Vedic texts, we find ample references to Gold, Silver, iron, copper, and their alloys⁶. Therefore, it is not surprising to find that mining, ore processing, extraction of metals, methods for metalworking, and alloying were known to them. In this section, we will see in more detail the above aspects of Indian metalworking and metallurgical principles⁷.

A detailed study of the archaeological remains and the repository of literary works from 5th century CE to 12th century CE, in addition to numerous references in the Vedic texts, clearly point to certain unique aspects of metalworking and metallurgy in India during ancient and medieval times. Some of them include the following:

- ♦ The smelters and metalsmiths had gained a high degree of knowledge regarding furnace design, combustion of fuel, refractories, and skill in operating the furnaces at the desired temperature. They were also aware of the temperature specifications and other conditions required to carry out the process.
- ♦ The knowledge of the Indian metal smiths on Iron-Carbon alloy was clearly superior and ahead of the times. This is evident from their ability to produce the famous wootz steel, which was in demand in the West.
- ♦ Indians were the first to introduce Zinc to human civilisation and also to develop Cu-Zn alloys.
- ♦ The Indian metal smiths were also adept in alloy technology as they could produce alloys of controlled composition.
- ♦ Indians developed good skills in designing and casting a variety of artefacts and deployed good moulding and diecasting methods. The available evidence suggests that they were casting small as well as large objects in the country.

10.3.1 Gold Extraction Process

The process for the extraction of gold in ancient times has been analysed and documented by Bharat Gold Mines Ltd. According to the available information, mercury was added to a mixture

- ♦ Hg, as well as HgS, have been extensively used in the preparation of Āyurvedic medicines.
- ♦ Hg is also used for the amalgamation process for the extraction of Silver and Gold.

of black sand and gold ore, and the mixture was rubbed with a little common salt. After some time, the gold and mercury would form an amalgam. The amalgam is separated by adding water to the mixture and agitating the mixture. The amalgam was rolled in a damp rag and the mercury was squeezed off. Finally, it was burnt in the fire to extract the gold. If the gold was found alloyed with silver or other base metals, it was hammered into a thin sheet between

two stones. The plate was burnt in a two-layered cow-dung. The cow dung absorbed the base metals, and the pure gold was separated. This shows that the technique of Hg (Mercury) amalgamation and gilding were apparently known to the Indian craftsmen from the very early times⁸.

However, the most common process of separation of native gold from the sand and quartz rock employs a simple gravity separation or panning. In this process, the quartz is first crushed to a fine size to extract the gold particle and then separate it by suspending the mixture in water. When the pan containing the suspension is agitated under water gold particles settle at the bottom of the pan and the sand and soil are washed away. In the case of a larger scale of operation, big pans were suspended in the river water through a sling so that the mixture could be easily agitated.

The main purpose of the extraction of gold was to make ornaments and coins. It was also used as 'Suvarṇa Bhasma' in Āyurvedic preparations. Ancient craftsmen could subject the purified gold metal bar to metalworking techniques such as forging, punching, embossing, etc. so that ornaments of intricate designs could be produced. It appears that ancient craftsmen could also use other metal forming techniques such as rolling and wire drawing. Gold coins were either made by die casting in clay moulds or by the technique of punch marking⁹.

10.3.2 Zinc Production

Of all the base metals, Zinc was perhaps the last to be discovered and used during the pre-Christian era. A study by Hegde, Craddock, and Sonawala in 1984 concluded that India was the first to introduce this metal to the rest of the world between 600 to 200 BCE¹⁰. Geographically Rajasthan stands out as a major mining province dating back to about 1000 BCE. Studies based on the debris and slag dump in the Zawar region of Rajasthan it is estimated that about 15,000 tonnes of Zinc ought to have been mined and processed. Zinc was exported to other countries even before the 11th century CE. Studies on the Zawar zinc mines provide rich insights into zinc mining in ancient India. According to some other estimate, about 250,000 tonnes of zinc concentrates ought to have been extracted from about 2.5 million tonnes of ore in the mined area, before modern mining operations commenced.

Extensive archaeological excavations conducted in several parts of Rajasthan clearly show that the earliest artefact containing an appreciable amount of zinc anywhere in the world is from India. A pioneering contribution of the Indian subcontinent in terms of the history of science and technology was the mastery of the metallurgy of zinc production. In the Aravalli ranges of northwestern India, amongst rich polymetallic mineral deposits one can see several mining galleries, slag heaps, and retorts which bear testimony to the early extraction of lead and silver at Dariba and Agucha and, most uniquely of zinc, at Zawar. The Zawar area near Udaipur has yielded unique evidence suggesting that the extraction of metallic zinc flourished on an almost semi-industrial scale by the mid-14th century CE as indicated by some carbon dates from the analysis of archaeo-metallurgical debris. Surveys and excavations revealed the extensive presence of furnaces or *koṣṭhis*, intended for an ingenious process of zinc extraction by downward distillation¹¹.

Zinc smelting was done during the 9th century CE, which was upgraded to an industrial scale around the 13th century CE. Several clay retorts, presumably used for zinc smelting are found scattered in the ruins of Zawar. At Zawar elements of transport systems are evident, with zig-zag paths on slopes with stone or wooden steps or ladders. The size of the passage on the main routes indicates that it would have been possible to carry loads in baskets on the head. Since India had plenty of iron from the early days, the tools used for extraction of the ore such as chisels, hammers, and scrappers were made of iron. For lighting, oil lamps were used, and the available evidence indicates that the lamps were placed at fixed positions in the mining areas.

Ancient Indians adopted a novel technique of downward drift reduction distillation process for Zinc production, which is a precursor to the modern processes adopted all over the world. The Zinc has a melting point of 410°C and a boiling point of 930°C. Moreover, in the open air at about 550°C Zinc gets oxidized to ZnO (Zinc Oxide). Therefore, the only viable method to extract Zinc is to rapidly cool the Zinc vapour to around 500°C to produce the liquid metal thereby preventing its reoxidation. The ingenuity of the ancient Indians was making this possible using the downward drift reduction distillation process. Some studies suggest that in India the process of distillation of water, wine, and probably mercury was known even in the Vedic period (Yajurveda). The distillation apparatus found near the ancient

- ♦ Ancient Indians adopted a novel technique of downward drift reduction distillation process for Zinc production, which is a precursor to the modern processes adopted all over the world.
- ♦ The distillation apparatus found near the ancient Takṣaśilā (Needham) was unique and much advanced.

Takṣaśilā (Needham) was unique and much advanced. This system uses a separate condenser tube fitted to the mouth of the distill to carry the vapours to the condenser vessel which was water-cooled. A similar apparatus called Damru-yantra was used for the preparation of Āyurvedic formulations. We will discuss different yantras used for such extraction purposes in Section 10.5.

The principle of the downward drift distillation process can be explained in simple terms using the description of a yantra available in Rasa-Ratna-Samuccaya (RRS) for extraction of



FIGURE 10.2 A Simple Illustration of the Yantra Used for Downward Drift Distillation Process

Source: Sen, S.N. and Subbarayappa, B.V. (2009). "A Concise History of Science in India", Indian National Science Academy, 2nd ed., p. 429. Reproduced with permission.

Zinc, Mercury, etc. Imagine having a pair of pots placed one over the other with their mouths aligned. What it means is that the lower pot is in the normal position and the upper pot is placed upside down over the lower pot. The upper inverted pot has a solid charge and is sealed with clay. It has a reed stick at the centre for the escape of gases. During the processing, the upper pot is heated by building a firing platform around it. Once it reaches 600°C, the reed is charred and burnt off, paving way for the reduced metal vapour to flow downwards. The lower pot acts as a condenser and it has mechanisms for rapid cooling so that it can convert the vapour into liquid metal. Figure 10.2 is a simple illustration of the yantra for the downward distillation process.

A similar but advanced method was used by the Zawar smelters. The process used by them consisted of the following steps. Zinc is extracted by smelting Zinc Sulphide ore. After sizing, the ore was mixed with charcoal dust and fired into a heap to convert it into oxide by

controlled roasting in an oxidizing atmosphere. During this process, ZnS (Zinc Sulphide), PbS (Lead-II Sulphide), and other sulphide minerals got converted to their respective oxide but no reduction of these oxides by carbon was permitted. The roasted ore was mixed with more charcoal powder, salt, and borax as flux and thoroughly mixed with cow dung and water, and then made into balls of 5 to 10 mm dia. by hand rolling. These pellets were dried in Sun and then filled into the brinjal-shaped retorts.

Many retorts have been found at Zawar. They are of two principal sizes with a capacity of 750 c.c. and 2000 c.c. As described in RRS, the main retort or crucible is in the shape of a brinjal. It was made from a clay mixture consisting of locally available refractory clay, rice husk, and sometimes mixed with iron ore dust. Many such compositions have been mentioned in ancient Indian literature.

10.3.3 Copper Mining and Extraction Process

India abounds in ancient specimens of copper. Utensils made of copper, owing to the peculiar sanctity attached to it by ancient Indians, have been used in India from ancient times in the

performance of religious ceremonies. Wires, copper, and brass have from early times been used in the construction of stringed musical instruments for which India has been famous. A long and flourishing tradition of the copper industry in ancient India can be inferred from the following archaeological specimens¹²:

- ◆ As many as 424 copper implements and weapons and 102 pieces of thin silver plates were discovered in one place measuring about three feet in length, three feet in breadth, and four feet in depth in the village of Gungeria in the Nagpur Division of the erstwhile Central Provinces in 1870 CE. The copper implements were mostly celts, shovels, axe-blades, spades, manufactured for warlike, domestic, and agricultural purposes. These have been kept in the Archaeological Department of the Indian Museum, Kolkata. Figure 10.3 is a sample of copper swords belonging to pre-historic times.
- ◆ A big solid copper bolt was found in the Rampurwa Ashoka pillar near the frontiers of the Kingdom of Nepal. The bolt is barrel shaped in appearance, slightly tapering at the two ends. It is 242 inches long, with a circumference of 14 inches at the centre. The two ends of the bolt have a circumference of about 12 inches. This bolt ought to have been fabricated as early as the third century BCE, thereby testifying to the high metallurgical skill of ancient Indians.
- ◆ A colossal copper statue of Buddha was discovered at Sultanganj in the district of Bhagalpur in the ruins of an old Buddhist monastery (see Figure 10.4). It was 7 feet 6 inches high and weighed nearly 1 ton. Based on the mode of its construction and discovery of a coin belonging to the time of Candragupta II in the vicinity of the monastery, its date is estimated to be 5th century CE. Lumps of copper ore were also found suggesting that the smelting and casting operations were done on the spot. It has been taken away and preserved in the Birmingham Museum.
- ◆ Another statue, 80 feet high was reportedly witnessed by the Chinese traveler Hieun-Tsang near Nālandā in a Vihāra of brass constructed by King Śilāditya (also known as King Harṣavardhan who ruled from 606–647 CE). According to Hieun-Tsang, "A pavilion of six stages is required to cover this gigantic copper. It was the work of Rāja Pūrṇavarman, the last descendant of King Aśoka, a king of 7th century CE".
- ◆ The principal use of copper was to mint coins by the kings. The punch-marked copper coins of many kings of Northern India (1st century BCE), copper coins of the Kanishka



FIGURE 10.3 Prehistoric Copper Swords Discovered at Fatehgarh (Indian Antiquary, 1905, p. 236).

- ◆ The Chinese traveller Hiuen Tsang has written that Indians knew the method of preparing brass from a mixture of Copper and Calamine.
- ◆ Rasa-ratna-samuccaya gives a vivid account of the processes of extraction of copper and its use in the preparation of various Āyurvedic formulations.

methodology using a process where the roasting and reduction is carried out in an open pit furnace. In such a furnace temperature as high as 1000°C can be easily obtained and the reduced metal flows out and gets collected in the front pit. At this temperature, Ag_2S (Silver Sulphide) also gets decomposed and reduced and forms an alloy with molten lead. Silver metal was subsequently obtained from the alloy of Pb-Ag by reheating it in a shallow hearth furnace with bone charcoal at the bottom. During remelting of the alloy, excess air is blown so that the lead gets oxidized into PbO . Since this has a very low melting point it is skimmed off, leaving behind the purified liquid silver. It is tapped out and cast into ingots¹⁷.

Kautilya's Arthaśāstra describes Silver-Copper alloys and the techniques applied for minting silver coins of a definite weight. The study of the punch marks on ancient coins indicates a high level of craftsmanship in the preparation of punching dies.

10.4 IRON AND STEEL IN INDIA

Ancient specimens of iron are widely available in India which will convince anyone that India has always been a rich iron-producing country. A host of archaeological excavations firmly establish that ancient Indians were well aware of the use of iron. Indians manufactured massive iron objects during the Christian era, which were much ahead of the times. The iron pillar at Qutub Minar (in the Mehrauli area of South Delhi) which weighs about 6,000 kg is well known, but other iron pillars, viz the one at Dhar (12th century CE) which weighs 7,000 kg and is nearly twice as big as the Iron Pillar of Delhi in Qutub Complex, and another on Mount Abu, which is less known. Another notable one is the 29 iron beams in the Konark temple in Odisha. Moreover, the gigantic iron beams at Konark, which lay buried in the sea sand for a few centuries were recently unearthed. These as well as numerous iron beams at Puri and Bhubaneswar where as many as 239 pieces were found in the Puri Gunduchibari temple are shining evidence for the supremacy of iron in ancient Indian culture¹⁸. The corrosion resistance and sheer size of these artefacts point to highly evolved practices in metalworking involving iron. Moreover, the presence of numerous Aśoka pillars of stone cut out in a faultless manner from single pieces of stone of gigantic dimensions presupposes the use of the finest steel saws and steel chisels in India in the 4th century BCE.

A large number of prehistoric iron implements such as swords, daggers, tridents, spears, javelins, arrows, spades, hangers, saucer lamps, beam rods, and tripods has been unearthed during the excavation of numerous burial sites in the Tuticorin district of Tamil Nadu. The stūpa of Bodh Gaya belongs to the Aśokan times. The foundations of the stūpa, on excavation, have a piece of iron slag that has been preserved in the Kolkata Museum. This piece of iron slag dated the 3rd century BCE is believed to be the most ancient archaeological evidence of the manufacture of iron in India. Besides the iron slag, many iron clamps five or six inches long and about one inch broad have been found in the main temple and in various stupas in Bodh Gaya, which are preserved in the Indian Museum¹⁹.

With the advent of the carburisation of iron, a special type of high carbon steel was produced in India from as early as the fourth century BCE, known as wootz steel, used for military applications for producing tough swords, helmets, and armour. The word 'wootz,' by which name Indian steel was and is still known in Europe seems to have been prepared from time immemorial and was the metal

- ◆ Indians manufactured massive iron objects during the Christian era, which were much ahead of the times.
- ◆ The Suśruta saṁhitā has mentioned the fabrication of more than 100 surgical tools made of iron-carbon alloys and the process of heat treatment to obtain a razor-sharp edge.

from which the famous Damascus blades were prepared. From the presence of numerous Aśoka pillars of stone neatly carved out of single pieces of stone of gigantic dimensions we can infer the use of saws and chisels made out of steel in India in the 4th century BCE. The use of a large number of surgical instruments by Suśruta (3rd century BCE), some of which 'could bisect a hair longitudinally,' also points to the use of steel in India in making cutlery and such medical instruments²⁰.

It appears that the ancient Indians, especially from the time of the Guptas, were aware of the technology of producing corrosion resistance iron. This calls for a good understanding of metallurgy and material science, to use modern terms. Dr. P.C. Ray in his *History of Hindu Chemistry* remarks, "...The wrought iron-pillar in Qutub complex, which weighs ten tons and is some 1500 years old, the huge iron girders at Puri, the ornamental gates of Somnath, and the feet wrought-iron gun at Nurwar are monuments of a by-gone art and bear silent but eloquent testimony to the marvellous metallurgical skill attained by the Hindus...."²¹ The high status of iron and steel technology in ancient and medieval India is reflected in the manufacture and use of numerous large iron objects, including forge-welded cannons. Such cannon, found at Nurwar, Mushirabad, Dhaka (in Bangladesh), Bishnupur, Bijapur, Gulbarga, and Thanjavur, bear ample evidence to the medieval Indian blacksmith's skill in the design, engineering, and construction of large forge-welded iron objects. Based on its weight and size, the cannon at Thanjavur (see Figure 10.5), must be regarded as one of the largest forge-welded iron cannons in the world. According to a recent authoritative history of the ancient city of Thanjavur, the cannon was manufactured in Thanjavur during the regime of Raghunatha Nayak (1600–1645 CE)²².



FIGURE 10.5 The Thanjavur Cannon

Source: By P. Jambulingam—Own work, CC BY-SA 4.0, <https://commons.wikimedia.org/w/index.php?curid=35437933>

The Delhi iron pillar (5th century CE) is made of almost pure iron (99.72%) and is crowned with the decorative capital. The pillar is unrivalled for it is corrosion-free for such a long time and withstood the ravages of weather (see Figure 10.6). The Sanskrit inscriptions found on the pillar indicate that it was constructed during the reign of Candragupta Vikramāditya II (375–414), of the Gupta dynasty. The total length of the pillar is 23 feet 6 inch, including the decorated capital which is 3 feet and 5 inches tall. It was reported that the top of the pillar

the introduction of the process of production of Wootz steel, which is a classic innovation of ancient Indians.

Yuktikalpataru an 11th century CE work mentions the relative properties of iron-carbon alloys produced in different regions of India and provides a comparative metric for their relative superiority as given below³²:

- ♦ Krauñca—iron is supposed to be two times better than Sāmānya (probably Muṇḍa-loha).
- ♦ Kaliṅga (Odisha)—8 times better than Crouñca iron.
- ♦ Bhadra—100 times better than Kaliṅga iron.
- ♦ Vajra—1000 times better than Bhadra iron.
- ♦ Pāṇḍi—6 times better than Vajra iron.
- ♦ Niravi—10 times better than Pāṇḍi iron.
- ♦ Kānta—Ten billion times as good as Niravi iron.

10.5 LOST WAX CASTING OF IDOLS AND ARTEFACTS

This is a process adopted for producing metal sculptures based on an original sculpture. The metal sculpture is made of silver, gold, brass, or bronze. The beautiful idols and icons made during the Chola era were produced using bronze using this technique. Figure 10.9 is an example of a bronze idol of Śiva-Pārvatī, made using the lost wax casting process probably in the 11th century CE. In the Indian tradition, the lost-wax casting process is also used to produce idols using a combination of five metals (Gold, Silver, Lead, Copper, and Zinc) known as pañca-loha idols. Intricate works and a good post-casting finish can be achieved by this method. Such is the specialty, antiquity, and grandeur of these pañca-loha idols that a good number of them have been stolen, smuggled, or simply taken out of the country. The oldest known example of this technique is a 6,000 year-old amulet from the Sindhu Sarasvati Civilisation. There are other examples from somewhat later periods from Mesopotamia in the third millennium BCE.

In this process, a pattern of the desired shape is made out of beeswax first (therefore the process is known as *Madhūcchiṣṭa Vidhānam*—meaning process using remains of beehives). A mould is then prepared by applying coatings of prepared clay slurry on it. Later, when the refractory clay slurry has dried the wax is removed by baking the refractory shell. Before casting the molten metal, the refractory shell is generally embedded in a box filled with the sand-clay mixture for further support to the refractory wall. After pouring the metal and its solidification the mould is carefully dismantled to take out the metal casting. Further refinements and finishing touches are given by filing, chiselling, engraving, and polishing.

Several literary sources in India have documented the lost wax metal casting process in detail. Chapter 14 of Viṣṇu-saṃhitā, which is a part of Viṣṇu Purāṇa, dated to 5th century CE mentions the need to make a wax model first before making a metal replica. Similarly, Chapter 68 of Mānasāra has details on *Madhūcchiṣṭa Vidhānam*. Manasollāsa and Abhilaṣitārthacintāmaṇi of 12th century CE also provide a detailed account of the method of preparation of wax pattern and slurry coating. As per these texts, the composition of the clay slurry coating consisted of clay mixed with a certain proportion of finely ground charred rice husk and sodium chloride. Some studies point to the use of the lost-wax technique currently by the tribals of Bastar for producing various brass and bronze objects.

For instance, RRS has described a variety of crucibles (Mūṣā) and furnaces for the manufacture of Āyurvedic medicines. It also describes in total 51 kinds of metallurgical tools (upakaraṇas), 36 kinds of equipments (yantras), 17 types of crucibles, and 9 types of furnaces. The alchemist, the Āyurvedic expert, and the metalworking engineer seem to be benefiting from each other's know-how and used similar apparatuses, albeit in varying sizes to suit their requirements. We have already seen some of these in the previous sections. However, we shall discuss this aspect in a focused manner here.

The crucible, known as mūṣā-yantra is the frequently used apparatus by the alchemists as it serves as a generic container for a variety of operations. The mūṣā-yantra was predominantly earthen. To prepare the crucibles, the earth of anthill, rice husk, iron rust, chalk, human hair, and a few other ingredients are mixed in a certain proportion, and to this goat's milk is added. The mixture is kneaded into a dough-like mass which would then be shaped into the desired forms, and finally sun-dried³³. There were various types of crucibles for different operations. For the extraction of zinc from calamine, a crucible of the shape of brinjal (Solanum melongena) is used. To this crucible, a tubular end is added, which expands towards its mouth like a flower.

We shall see in brief some of the apparatuses found in the ancient Indian laboratory³⁴. The apparatus is simple and made of earthen material shaped into different forms. In all the cases the source of heat was either cow-dung or wood. As we know, this may not yield a high temperature, and therefore wherever required the substances are heated for a few days to in some cases weeks.

Dolā-yantra uses a suspension mechanism using which the ingredient to be subjected to some treatment is kept in a piece of cloth, tied, and suspended using a rod. This is immersed in a pot half-filled with the desired liquid. The liquid is then heated from the outside. See Figure 10.10(a) for an illustrative sketch of Dolā-yantra.

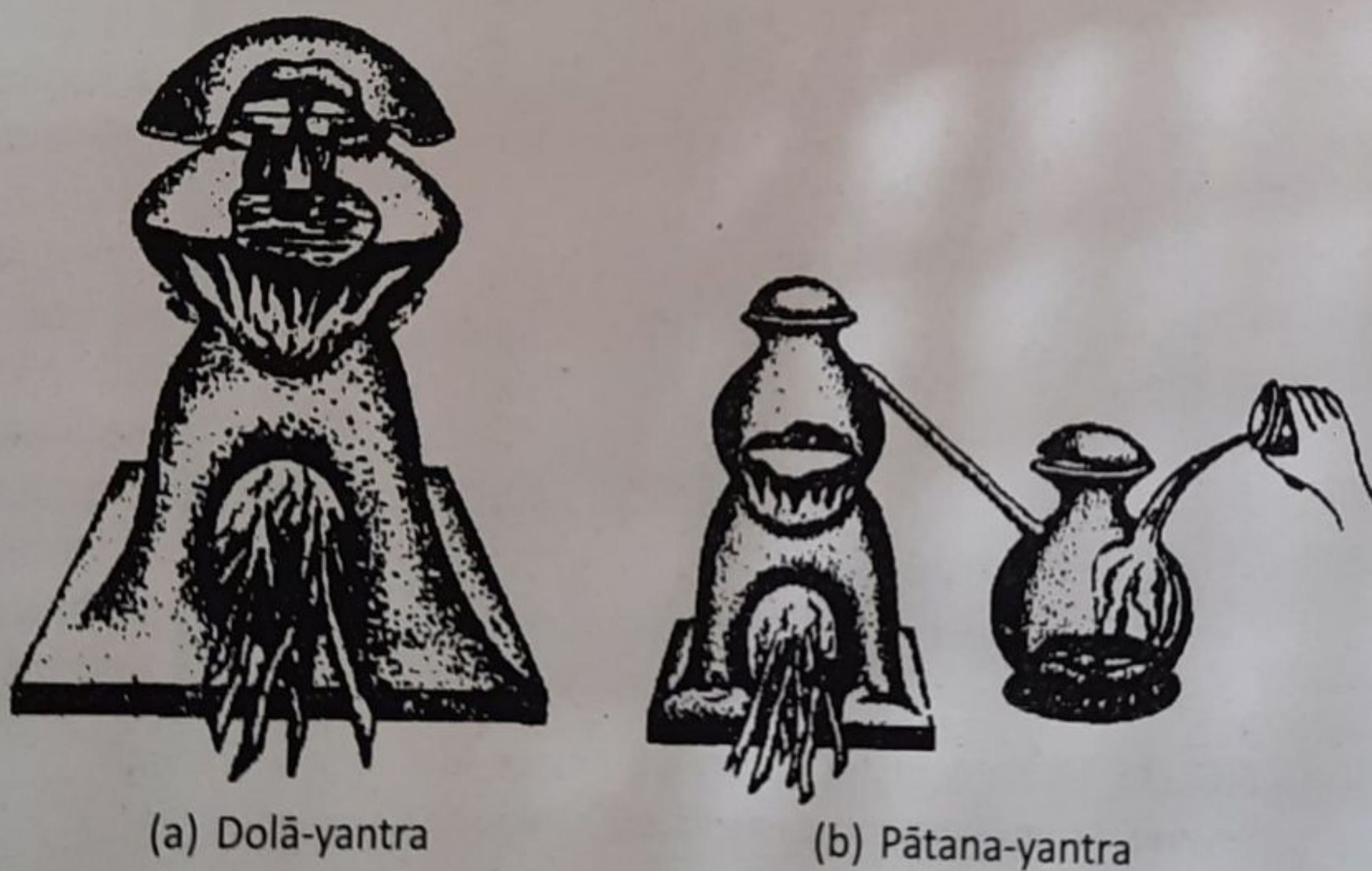


FIGURE 10.10 Examples of Yantras Used by Alchemists in India

Source: Sen, S.N. and Subbarayappa, B.V. (2009). *A Concise History of Science in India*, Indian National Science Academy, 2nd ed., pp. 426–430. Reproduced with permission.

Svedanī-yantra is used for steaming purposes. The mouth of a pot is covered with a piece of cloth and the substance to be steamed is placed on the cloth. Water is kept in the pot and is covered using another pot. The substance is steamed by boiling the water.

Pātana-yantra is used whenever a sublimation or distillation process is involved. It consists of a well-baked pot of suitable dimensions. Water is filled in this pot to its neck and another bigger pot is kept over it in an inverted position. The two necks of the pots are firmly sealed using an adamantine made of lime, raw sugar, and buffalo's milk. A downward sublimation process is applied as follows. When the substance is smeared inside the upper vessel and heated from outside by burning cow-dung cakes, it vaporizes and collects in the water in the lower pot. On the other hand, the substance may also be heated in the lower pot and the upper one is kept cooled by applying wet rags. The sublimate deposits in the interior of the upper pot. Figure 10.10(b) is a schematic representation of a version of Pātana-yantra for the downward distillation process. In this version, the vapour passes through a pipe into another pot kept alongside. The second pot is cooled with water and the vapour condenses into liquid and is collected in the second pot.

Dhekī-yantra is also used for the distillation of substances such as mercury. A hole is made slightly below the neck of a pot and one end of a bamboo tube is introduced into it. The other end is fitted into a brass vessel that is filled with water. Mercury mixed with the other desired substances is subjected to distillation in this type of apparatus.

Vālukā-yantra is of a type of sand-bath for heating substances uniformly and for usually a long time. A long-necked bottle, containing the substance to be heated, is kept buried in sand in an earthen pot up to three-fourth of its height. The apparatus is heated from below for a long time till a straw when placed on the surface of the sand, catches fire. Instead of sand, sometimes salt is used.

- ♦ Yuktikalpataru an 11th century CE work mentions the relative properties of iron-carbon alloys produced in different regions of India.
- ♦ The alchemist, the Āyurvedic expert, and the metalworking engineer seem to be benefiting from each other's know-how.

Dhūpa-yantra is used for fumigation purposes. Fumigation of the leaves of gold or silver with the fumes of sulphur or arsenic substances is done in this apparatus. Two vessels are employed for this purpose. In the lower vessel, iron bars are placed in a slanting position below its mouth, and leaves of gold or silver are placed on them. Sulphur or the arsenic substance is placed in the lower vessel and the other vessel is used for covering purposes. On heating, the fumigation takes place.

For heating mercury and sulphur together, the following details of an mūṣā-yantra are given in the Rasārṇava³⁵: The apparatus consists of two crucibles; one of them has a narrow orifice. In one crucible sulphur is taken and in the other mercury. Mercury and sulphur are to be moistened with filtered garlic juice. The crucible containing sulphur is to be inserted into the one having mercury, and the apparatus carefully lowered into an earthen pot over which another earthen pot is to be kept, and the rims luted with a cloth. It is heated from outside by cow-dung fire for three days.

In Chapter 10, we discussed how a culture of Science, Engineering, and Technology ought to have been part of the ancient Indian society based on archaeological excavations, and remains of metalworking mines and furnaces that were in use long ago. On account of appropriate know-how, it has been possible to extract large quantities of metals from the oars, manufacture corrosion-resistant steels that can stand the test of time, build lofty physical structures of staggering magnitude such as temples, deploy irrigation and water management systems, etc. In the area of health, we find instructions and processes pertaining to surgical practices, preparation of therapeutic formulations involving metallic compounds and their extracts. Ancient Indian literature also points to maritime trade between several parts of peninsular India with the countries in the far east, China, and the Roman Empire. This presupposes knowledge of building ships that are capable of withstanding long voyages across the oceans.

The book on Indian Science and Technology in the 18th century by Dharampal provides interesting insights and suggests that various Indian technologies were in a flourishing state in the 18th century CE. Several contemporary European observers, commented on the sophistication, efficacy and possible adoption in the Europe of the Indian science and technology practices. There are detailed descriptions in the book on widespread practice of inoculation against smallpox, practice of plastic surgery which was transmitted from India to the Europe and the quality of Madras Mortar to name a few¹.

In this chapter, we shall explore the ancient and medieval Indian literature for specific process details and methodologies for other aspects of engineering and technology. It remains an unanswered question as to what happened to these skills and working methodologies that ancient Indians seem to have practiced over several years.

11.1 LITERARY SOURCES FOR SCIENCE AND TECHNOLOGY

As we have seen in some of the earlier chapters (such as Mathematics and Astronomy) there has been a continuous tradition of scientific inquiry in the ancient Indian civilisation for more than 2,000 years. Therefore, it will be possible for us to locate some literary evidence for Science and Technology (S & T) in ancient Indian works, which provides another source for us to make certain conclusions about S & T thinking. Table 11.1 has a list of literary sources that discuss S & T principles and applications. Let us briefly see some areas of applications:

Metallurgy, Material Working, Mining: Fabrication of iron pillars that withstands more than 1000 years of corrosion does not happen by mere accident. It must be preceded by a good knowledge of metallurgy and the availability of experienced ironsmiths. Similar evidence in copper, zinc, brass, and bronze point to the availability of know-how. While references to metals are found in the Vedic corpus, some treatises pertaining to the 5th century CE to 13th century CE discuss these issues in some detail.

The Vedic corpus has multiple references to iron (Ayas), copper (Loha or Lohitāyas), gold (Svarṇa), silver (Rajata), and tin². For instance, the picture of a blacksmith melting iron in his forge is given in R̥gveda 6.3.4³. In Chāndogya-upaniṣad 4.17.7 there is a mention of joining of metals gold, silver, tin, and lead⁴; In Atharvaveda 6.141.2⁵, a knife made of 'red metal' has been mentioned. In addition to the word *lohita*, the modern equivalent of copper, viz. *tāmra* occurs in one passage (Atharva, 10.2.11)⁶. Several Indian texts such as Rasaratnākara, Rasāyana-śāstra, and Rasaratna-samuccaya deal extensively on the zinc metallurgy. The Mānasōllāsa, written by

the Chalukyan king Someśvara of the 12th century CE describes the process of making metal icons (Idols of Gods for the temples) by the lost wax casting process. This is still practiced in Swamimalai in the Thanjavur district, where there are some world-renowned staphys and metalsmiths engaged in idol making. In Caraka-saṃhitā references of zinc, and its extraction from ore is available. Rasaratna-samuccaya of Vāgbhaṭa, is considered to be a metallurgical encyclopaedia of the 13th century CE. There are vivid descriptions and procedures related to mineral purification; extraction of metals; fuels, crucibles, furnaces; metals, alloys, and their properties.

TABLE 11.1 S & T References in Ancient Indian Literature

No.	Period	Source	Issues of S & T Discussed in the Text
1	Up to 1st Century BCE	R̥gveda	References to Iron, ironsmith and melting process in several passages – no mention of copper
		Atharvaveda, Yajurveda	Mention of six metals (gold, silver, iron, tin, lead, copper), various technical occupational categories
		Chāndogya-upaniṣad, Taittirīya-brāhmaṇa	Mention of black metal and red metal
		Rāmāyaṇa	Description of iron ores, metals and mines
		Manusmṛti	Mention of the use of household utensils made of copper, iron, bronze, brass, tin and lead and enjoins their purification with ashes, acid and water
		Caraka and Suśruta Saṃhitās	Description of surgical instruments, mention of six metals, alloys of copper (brass, zinc), tools for surgical interventions
		Arthaśāstra	Primarily a book on statecraft and public administration, description of various warfare implements such as iron swords, arrows, axes, spades, etc., ores and mines of gold, silver, copper, iron, lead, tin and precious stones, state functionaries such as superintendents of mines, ships and weaving, use of copper in making alloys and in gold and silver coins, goldsmith roles
2	1st Century CE to 10th Century CE	Rasaratnākara of Nāgārjuna	Reduction of metals, recipes for transmutation of base metals, yantras for carrying out several physio-chemical processes, alchemical processes and preparations of mercurial compounds
		Bṛhat-saṃhitā of Varāhamihira	Primarily an astronomy treatise, Geological formations (rain, dust storm, water, etc.) mining, earthquakes, yantras, medicine, botany, architecture (temple, idols, etc.)
		Amara-kośa	Technical terms, vocabulary

(Contd.)

TABLE 11.1 S & T References in Ancient Indian Literature

No.	Period	Source	Issues of S & T Discussed in the Text
3	11th Century CE to 14th Century CE	Yukti-kalpataru by Bhojarāja	Architectural principles, ideas of different types of ships and shipbuilding
		Samarāṅgaṇa-sūtradhāra of Mahārājādhirāja Bhojadeva	Architectural principles, different types of yantras (devices), ideas of aerial vehicles
		Mānasollāsa of king Sōmeśvara	Process of making metal icons by the lost wax casting process
		Rasacintāmaṇi of Madanāntadeva	Preparation of medicine, information on chemistry and metallurgy, including the production of zinc
		Rasaratna-samuccaya of Vāgbhaṭa	Contemporary method of production of zinc, different categories of iron, mineral purification; extraction of metals and mercury; fuels, crucibles, furnaces; metals, alloys and their properties, 51 metallurgical implements, 36 types of equipment, 17 types of crucibles, and 9 types of furnaces. It is considered as a metallurgical encyclopaedia of the 13th century, pharmaceutical procedures of metals and minerals, description of many herbo-metallic and mineral formulations

Alchemy: Metallurgy is closely related to alchemy, which is a science of transmutation of metals. Alchemy was practiced in ancient civilisations to find methods to transform other metals into gold. In the Indian tradition, we see similar discussions. Nāgārjuna, a well-known Indian alchemist was a Buddhist monk who authored Rasaratnākara. The Rasaratnākara gives several recipes for the transmutation of base metals into silver and gold, the core subject matter of alchemy. It also describes many chemical processes like the extraction of zinc, mercury, and copper. The text also mentions the names of several yantras (apparatus) for carrying out various Physico-chemical processes like distillation, sublimation, extraction, and calcination. Rasendra-cūḍāmaṇi by Somadeva and Rasaprakāśa-sudhākara by Yaśodhara are the other important treatises that deal largely with chemistry and alchemy. A major work, Rasaratna-samuccaya of Vāgbhaṭa extended the application of alchemy into the health and treatment of diseases using heavy metals.

Architecture and Town Planning: The presence of a large number of temples throughout the country points to good knowledge of architecture, iconography, building construction, and construction management practices. Many works address these issues, and we present a list of important literary works in Chapter 12. However, we merely mention two of them here for some other S & T applications.

Samarāṅgaṇa-sūtra-dhāra of Mahārājādhirāja Śrī Bhojadeva of Dhārā, who lived in the 11th century CE is primarily a treatise on architecture (vāstu-śāstra). However, in Chapter 31 of the work, there is a description of different yantras (man-made devices) that can be put to different uses. Further, there is a description of aerial vehicles, which can be considered a

good pre-cursor to current-day aerial vehicles. Another interesting work Yukti-kalpataru, by one Bhojarāja, is also considered to be of the same period. There is no clarity as to whether both are by the same author. This work is also mainly on architecture (vāstu-śāstra). There are descriptions in this work on types of ships, shipbuilding, etc.

Medical Applications: Caraka-saṃhitā and Suśruta-saṃhitā are the two major medical treatises of ancient India. In Suśruta-saṃhitā, one finds descriptions of surgical instruments for performing several surgical operations. Copper (and its alloy brass and bronze), iron, and other metals including zinc, gold, and silver find mention in both the treatises.

Other Engineering Applications: The Bṛhat-saṃhitā of Varāhamihira is primarily an encyclopaedic work, covering mainly astronomy. However, there are other interesting issues discussed in the work, which include geological formations such as wind patterns, rain formation, earthquakes, and water tables. It also discusses certain aspects related to medicine, yantras, and botany. The lexicographical work of Amarakośa provides the required vocabulary for the engineering aspects, metals, and various physical structures and thus points to the existence of these in the society during that time.

11.2 PHYSICAL STRUCTURES IN INDIA

While archaeological evidence exists throughout the length and breadth of the country, we know very little about the scientific ideas of the people of the ancient civilisation. One possible reason for this could be the reported large-scale burning of over six million manuscripts kept in Nalanda University by the invaders. No civilisation can weather such catastrophic damage inflicted all of a sudden. It is bound to create a certain discontinuity in the knowledge tradition. However, the Indian civilisation is a 'living tradition' and we have a rich heritage of practices and systems that continue till date despite periodic disturbances. There are several ancient physical entities still in use or in a preserved condition. In particular, in India, there are several temples built over the last 2500 years, still either in active use by society or preserved as monuments. These can offer valuable insights into the status of S & T in India. Therefore, one approach to assessing if there were any S & T heritage and knowledge among the people of ancient India is to examine the living specimens of yesteryears. By studying some aspects of ancient buildings and architectural monuments one can make some inferences about the S & T knowledge that ought to have gone behind creating these structures.

The footprints of S & T left behind by ancient Indians in the form of monuments and living physical specimens is too large to be meaningfully analysed and presented in a concise form. It is not even clear if any systematic and scientific study of these has been carried out. Instead, we need to take a slightly different and concise approach. A useful starting point is the UNESCO-approved World heritage sites in India. UNESCO designates certain places as World Heritage as they may have outstanding value to humanity. These could be 'cultural entities' such as ancient ruins, historical structures, buildings, and even cities or 'natural sites' such as rivers, and forests. The cultural sites are often considered to be a creative genius of human beings. We can explore the world heritage sites of the 'cultural' category in India for several reasons. The advantage of this is that these sites represent the developments in culture and social systems, and accumulated knowledge and skills of the society at that time and serve as an outstanding example of architecture, or technology. These sites also portray the ideas, living traditions, and artistic and literary works of the times when they were created. Of the 36

world heritage sites in India, a vast majority of them are cultural sites. Table 11.2 has a list of approved world heritage sites in India up to 13th century CE as of September 2020.

TABLE 11.2 World Heritage Sites in India (up to 13th Century CE)

Sl. No.	World Heritage Site	Period	Short Description of the Site and its Importance
1	Rock Shelters of Bhimbetka, Madhya Pradesh	30,000 years	Five clusters of rock shelters with paintings in some of the shelters dated from 100,000 BCE to 1000 CE amidst dense forest has been discovered. Twenty-one villages surrounding them reflect the traditions displayed in the rock paintings.
2	Champaner-Pavagadh Archaeological Park, Gujarat	Prehistoric and 8th–14th Century CE	Largely unexcavated archaeological, historic and living cultural heritage properties which includes prehistoric sites, a hill fortress of an early Hindu capital, and remains of the 16th-century capital of the state of Gujarat.
3	Mahābodhi Temple Complex at Bodh Gaya, Bihar	3rd Century BCE to 6th Century CE	The first temple was built by Emperor Aśoka in 260 BCE around the Bodhi Tree. The oldest temple in the Indian sub-continent built during the Gupta period.
4	Sanchi, Madhya Pradesh	2nd Century BCE to 12th Century CE	A group of Buddhist monuments dated between 200 BCE and 100 BCE. The sanctuary has a plethora of monolithic pillars, palaces, temples and monasteries in different status of preservation.
5	Ajanta Caves, Maharashtra	6th–2nd Century CE	31 rock-cut cave monuments depict richly decorated fresco paintings, which are unique representations of the religious art of Buddhism.
6	Archaeological Site of Nalanda Mahāvihāra at Nalanda, Bihar	5th–12th Century CE	Archaeological remains of a monastic and scholastic institution. It includes stūpas, shrines, and vihāras (residential and educational buildings). The most ancient university of the Indian Subcontinent that functioned uninterrupted for a period of 800 years.
7	Elephanta Caves, Maharashtra	5th–8th Century CE	Consists of two groups of caves – the first is a large group of five Hindu caves containing rock cut stone sculptures, the second, a smaller group of two Buddhist caves.
8	Ellora Caves, Maharashtra	600 CE to 1000 CE	A mix of artistic creation of the ancient civilization of India depicting Buddhism, Hinduism and Jainism. 34 monasteries and temples sculpted contiguously into rock walls.
9	Mahabalipuram, Tamil Nadu	7th and 8th Century CE	About 40 monuments including the largest open-air bas-relief in the world carved out of rock.
10	Hill Forts of Rajasthan	7th–16th Century CE	Rajput forts at Chittor, Kumbhalgarh, Ranthambore, Gagron, Amer and Jaisalmer characterized by its mountain peak settings, utilising the defensive properties of the terrain.
11	Pattadakal, Karnataka	8th Century CE	Nine temples representing a fusion of Nāgara and Drāviḍa architectural features, as well as a Jain sanctuary. The Virūpākṣa Temple, is an outstanding architectural edifice.

walls of the temples. With many sculptures of varied themes, the Khajuraho temples vibrate with the exuberance of human warmth, not generally found in other such art forms.

The existence of musical pillars in some of the temples informs us that ancient Indians had a good understanding of the acoustical properties of stones. Above all, building large granite temples (such as the living temples of Cholas in Tamil Nadu) demands a good understanding of sourcing, logistics, and supply chain management. Moreover, knowledge of cutting stones using special tools is inevitable as granite belongs to the category of the hardest substances to work with.

11.3 IRRIGATION AND WATER MANAGEMENT

Hydrology and water management systems are extremely crucial for any agrarian-based economy. Expectedly, there is a long history of developing good irrigation and waste management systems in India. Several archaeological excavations and some of the living dams and anaicuts provide a wealth of information for one to infer these skills of ancient Indians. Several studies in the past have clearly brought out the presence of sophisticated water management systems in Dholavira, an important city in the Sindhu Sarasvati civilisation⁹. The city was surrounded by a series of 16 large reservoirs (7 m deep and 79 m long), with some of them interconnected together, which nearly accounted for about 10% of the area of the city. Recently, a rectangular stepwell was found at Dholavira measuring 73.4 m long, 29.3 m wide, and 10 m deep, making it three times bigger than the Great Bath of Mohenjo-Daro.

There was a survey conducted in an ancient Indian irrigation system at Sanchi, a well-known UNESCO World Heritage site in Madhya Pradesh, and four other known Buddhist sites of Morel-Khurd, Sonari, Satdhara, and Andher, all established between 300 and 200 BCE. The heights of the dams ranged from 1 metre to 6 metres and their lengths from 80 to 1400 metres. Two of the larger dams were equipped with spillways suggesting that flood protection was also taken into account while designing these structures. The Sanchi dams and another well-known Sudarsana dam in Gujarat had several similarities.

Emperor Candragupta Maurya is reported to have constructed the Sudarśana dam in Girnar, Junagadh, Gujarat to which some structural improvements were made during the reign of Aśoka. In an excavation work conducted by the Archaeological Survey of India during 1951–1955, a canal 45 feet broad and 10 feet deep was traced up to a length of 450 feet in Kumhrar (the site of ancient Pāṭaliputra), a few miles south of Patna, Bihar, which is attributed to the Mauryan period. The canal was linked with the 'Sone River' and also with the 'Ganges'. While excavating the area around the 'Heliodorus' pillar in Vidisa (present-day Vidisha, Madhya Pradesh), the remains of a 300 BCE canal were discovered.

Mauryans had developed a sophisticated concept for rainwater harvesting and irrigation management, known as the *Ahar-Pyne* system, which is still practiced in the regions of southern Bihar and Chhota Nagpur. The Pynes are constructed channels to utilise the river water flowing through the hilly regions, whereas the Ahars are catchments with embankments to store rainwater and the water released from the Pynes. The Pynes feed many Ahars and several distributaries are then constructed for irrigating the field. The Ahar-Pyne system also works as a flood mitigation system. The Government of Bihar has taken up renovation of the traditional water bodies (Ahar-Pyne system) under the *Jal Jeevan Hariyali* program in 2020. This shows how robust and useful is this ancient system for water harvesting even in modern times.

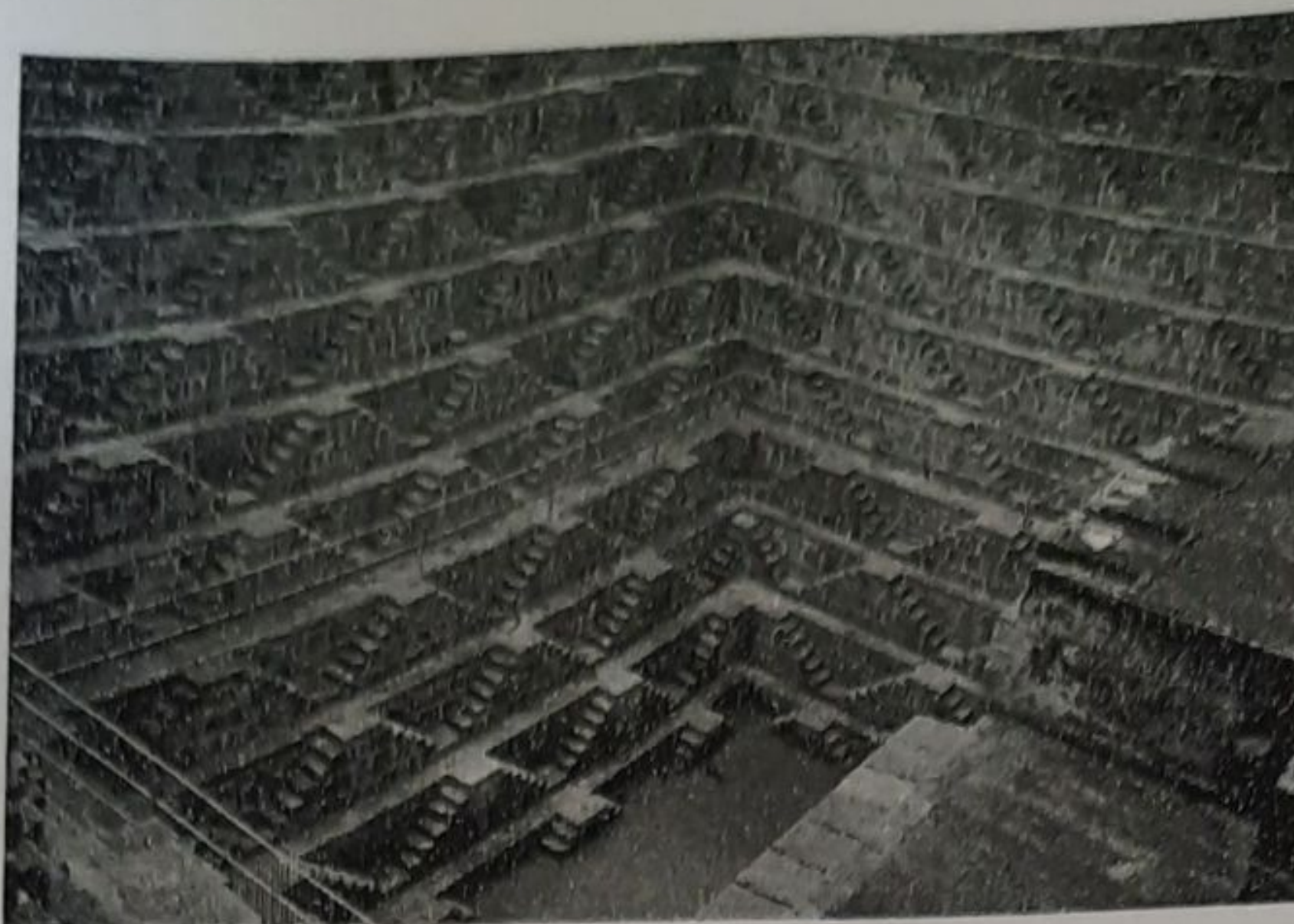


FIGURE 11.3 Chand Baori, Rajasthan, One of the Deepest and Largest Stepwells in India

Source: By Doron, <https://commons.wikimedia.org/wiki/File:ChandBaori.jpg>

11.4 DYES AND PAINTING TECHNOLOGY

Human beings essentially love colours. Adding colour to life both in a figurative and real sense is a long-time pre-occupation of human beings. People in ancient India had knowledge of the properties of different colouring agents and the technique of using dyes. They were using both organic and inorganic substances for these purposes. Dyeing wool or silk is easier since they have both basic and acidic properties and are proteins. On the other hand, dyeing cotton is difficult and it requires substances known as mordants, capable of penetrating and staying in the fibres. Pigments on the other hand are colouring agents. The use of these substances is also found in the ancient Indian society¹².

The spinning of wool and cotton has been common in the Indus valley civilisation. The discovery of a fragment of a dyed fabric at Mohenjo-Daro indicates that the Indus valley people were aware of the dyeing and the need to use mordants on cotton fabrics. Many dyeing vats were also found, confirming the fact that they had the know-how for dyeing fabrics. There are several references in our ancient texts about dyed fabrics with different colours and geometrical designs. These further add credence to the argument that ancient Indians were aware of the use of mordents for dyeing difficult fabrics such as cotton.

In ancient texts there is a frequent mention of certain colours. This includes Indigo (Nīla), Turmeric (Haridrā), Maddar (Mañjiṣṭhā) and Resin (Rajana). For instance, Krishna is supposed to have been draped in a yellow colour dress, while his brother Balarama is fond of blue colour robes. In Buddhist literature, the Vinaya texts describe six sources of dyes: roots, trunks, barks, leaves, flowers, and fruits of trees. Ancient Indians knew the art of selecting colours and mixing them to achieve different shades. Yaśodhara in his Jayamaṅgalā commentary on Vātsyāyana's Kāmasūtra has listed six components of the art of painting amongst which mixing of colours (varṇikā-bhaṅga) is one¹³. According to Viṣṇu-dharmottara-purāṇa, most common colours were white, red, yellow, black and blue. Several hundred other colours and shades may be obtained by a judicious combination of one or more of these basic colours (which a good artist must know)¹⁴. This suggests that the art of painting was well known in ancient and medieval India and that people knew the technique of preparing pigments, mixing colours to get shades of their choice for painting beautiful pictures.

IKS IN ACTION 11.2

The Art of Making Perfumes

If we read several ancient Indian texts, we find mention of use of perfumes, especially by the royal people. This suggests that ancient Indians knew how to use perfumes and methods of preparing such perfumes. Viṣṇu-dharmottara-purāṇa is one of the Upa-purāṇas and it consists of three kāṇḍas. The 64th chapter in the second kāṇḍa has 46 verses in which the issue of perfumes is discussed. The chapter also discusses the technique of preparing different kinds of incenses, perfumed water for bath and scented oil. The process of preparing the perfumes is described in eight steps (known as karmāṣṭakam):

1. Śodhanam—This involves purification of the ingredients
2. Vāsanam—Scenting with the perfumes of flowers
3. Virecanam—The process of cleansing
4. Bhāvanā—Saturation of powder in fluid
5. Pāka—Ripening by preparing a decoction
6. Bodhanam—Revival of scents using certain reviving agents
7. Dhūpanam—Fumigating with perfumed vapours
8. Vedhanam—Further process of revival

Another interesting source for perfumes is Varāhamihira's Bṛhatsaṃhitā (5th century CE).

In Chapter 77, titled Gandha-yukti, addresses the issue of preparation of perfumes in 37 verses. In the first 12 verses of the chapter, different methods of preparing perfumes using various ingredients are discussed. However, an equally interesting aspect of this chapter is a nice introduction to permutations and combinations. Varāhamihira in the rest of the chapter shows how using 16 ingredients it is possible to blend them in different ways.

If sets of four ingredients are to be made from the 16 ingredients, it results in $16C_4$ ($=1,820$) combinations. Further if the 16 ingredients are put into 4 sets of four each and in each of them the proportion of the ingredient are 4, 3, 2, and 1 respectively, then there are 96 permutations possible. This means that with the given instructions for perfume making it will be possible to prepare 174,720 ($=1,820 \times 96$) different perfumes.

Source: Sastri, V.S. and Bhat, M.R. (1946). "Varahamihira's Brihatsamhita", Soobbiah & Sons, Bengaluru, pp. 603–612 and Vyas, K.T. (1997). "Cosmetics and Perfumes", Chapter 4 [4.2] in *History of Technology in India*, Vol. I, A.K. Bag (Ed.), Indian National Science Academy, New Delhi, pp. 291.

Since all dyes are not soluble in water, they have to be rendered water-soluble by some reduction technique. Indigo has been popular since the pre-Christian era and was an article of export to Egypt and other countries from India. It is an insoluble dye, and some reduction techniques were applied to make it soluble. Thereafter, the cotton fibre was impregnated with the dye and exposed to air for oxidation. Alum, iron sulphate, rind of the lemon, cotton flower, lemon, green dried mangoes, and myrobalans were used as mordants for dyeing.

Dyes were often used in cosmetics. In ancient India, a paste of sandalwood used as cosmetics is often coloured with lac and other dyes, smeared over the body, and applied in different patterns. Collyrium (eye-salve) or añjana made of dark powdered antimony was very popular. Vermilion (sindūra), lac (lākṣā), and a yellow pigment gorocana were used for marking as tilaka on the forehead. Methods of beautifying the body using dyes are mentioned in Nāṭya-śāstra¹⁵. In Rasa-ratnākara, a few recipes are given for the preparation of kuṅkuma (used as tilaka). The boiled essence of flowers of the Palāśa tree (also known as the Indian Tulip) should be kept in the Sun in a wide vessel. Finely ground rice powder must be added to this in the proportion of 1:20, making it a fine paste. Further, the lime of half the volume should be mixed and stirred well in bright sunshine. Out of this, pellets can be prepared which will have a bright red colour like kuṅkuma.

In the Ajanta caves, the paintings in the 9th and 10th caves date back to the 1st or 2nd century BCE. Gupta period paintings are found in other caves of Ajanta, Badami, Chittannavasal, and temples of Kanchi and Thanjavur in Tamil Nadu. Ellorā mural paintings and Tanjore paintings of the 11th century CE beautifully depict figures in attractive bright colours and tints. At Ajanta, Badami and other caves, the colours most freely used were white, red, and brown in various tints, dull green, and also blue. These colours were obtained from the available substances. For example, lime sulphate was used for white colour. Using compounds of iron, they obtained red and brown tints. For yellow, the ancient painters used orpiment, a natural arsenic sulphide. The fresco paintings testify to the knowledge and skill of the painters in preparing permanent colours out of vegetable and mineral sources.

The Ajanta paintings employed extracts of plants and minerals in preparing pigments. First, the wall of the caves was given a pre-treatment using a special coating. Viṣṇu-dharmottara-purāṇa discusses different aspects of painting in the third kāṇḍa. In Chapter 40, it gives a formulation for preparing a coating that could be applied first on the surface (wall) where painting has to be done¹⁶. The ingredients of the coating

- ♦ Ancient Indians knew the art of selecting colours and mixing them to achieve different shades.
- ♦ The Ajanta paintings employed extracts of plants and minerals in preparing pigments.
- ♦ Viṣṇu-dharmottara-purāṇa discusses different aspects of painting in the third kāṇḍa.

include three varieties of brick powders mixed with one-third of clay. The other ingredients added to this include gum-resin (guggulu), beeswax (madhūcchiṣṭam), honey (madhu), a type of grass (kundaraka), molasses (guḍam), and safflower (kusumbha) soaked in oil, in equal proportions. Further powdered lime three-fourths burnt with bel fruit pulp and lampblack should also be added. After adding one-fourth part of the sand, the mixture should be soaked in water for a month to make it lubricous. With this paste,

a smooth and firm coat is applied to the wall. It is stated at the end of this procedure that preparing a wall in this manner will ensure that the painting does not get destroyed even after hundred years.

Preparing a Wall for Painting

The process of preparing adamantine for application on a wall as a coating in preparation for painting is described in the text Śivatattva-ratnākara. The process is as follows¹⁷. Fresh buffalo hide is to be boiled with water until it becomes a sticky butter-like paste. This is cut into small pieces and dried till the pieces become hard. This is the basic formulation for adamantine (known as Vajralepa) and is useful for preparing the initial coating on the wall to make it smooth and amenable for painting pictures. Sometimes it is required to add some pigments so that can be used as paints. The pieces of Vajralepa can be placed in a mud vessel filled with water and heated so that the pieces can melt. The molten liquid form of Vajralepa can be added to any colouring pigment to get the desired shade of a colour. Soft white sand and conch powder can be added to this Vajralepa and smeared on the surface of a wall three times so that it becomes smooth and glossy. This is followed by a coating using a mixture of Vajralepa and a moon-white mineral called Naga found Nilgiris. The wall finally is ready for painting pictures.

11.5 SURGICAL TECHNIQUES

Surgical methods, known as Śalya-tantra is an ancient technique known to Indians right from the pre-Christian era¹⁸. The most dangerous foreign object that required immediate

intervention is an arrow, with which the kings and generals were attacked in a battle. The surgical procedures were developed initially to attend to the wounded and remove the foreign objects from the human body thereby saving lives. Foreign objects of every kind are generally referred to as Śalya, but it specifically also means arrow. Śalya-tantra is, therefore, one of the oldest branches of Indian medicine.

In modern surgery, the use of anesthesia, hemostatic, antiseptics, and antibiotics are very important. It is still a matter of speculation as to how ancient Indian surgeons performed surgeries. The knowledge of vital points of the body (known as marma) was known to ancient Indians as was the case probably with Chinese and other older civilisations. This provided some sense of the anatomy of the body, which is required for surgery. Moreover, based on the techniques developed for pre and post-surgery in the Indian system, it is clear that the surgeons were aware of the critical nature of the infections that may develop post-surgery and the need for good post-surgical care. Suśruta is considered the father of Indian surgery. In his work Suśruta-saṃhitā, he has provided a clinical approach to some major surgical operations. Further, the other contributions of Suśruta include an emphasis on the training of surgeons, a meticulous code of personal ethics, and the social conduct of the medical profession in general. These topics will be of interest to today's surgeons and may help serve humanity better.

- ♦ Suśruta is considered the father of Indian surgery.
- ♦ In Suśruta saṃhitā, he has provided a clinical approach to some major surgical operations.

The status and quality of surgery are best evaluated by the basket of tools and instruments that a surgeon possesses. Suśruta's treatise makes a mention of 101 blunt and 20 sharp instruments. This included varieties of needles (straight, curved, or round body, blunt and cutting), suture materials, splints, and fracture beds for immobilisation of fractures at various sites, fourteen different types of bandages, lion forceps (Siṃha-mukha-yantra), and finger knife (mudrikā-śastra). Figure 11.4 has a sample set of sharp instruments described in Suśruta-saṃhitā. There is also mention of para-surgical measures, i.e. cautery, etc. for checking the bleeding. Suśruta has further mentioned the specific use of some instruments needed for the specific surgical manoeuvre, such as endoscopy to visualise the haemorrhoids and for conducting blind surgical procedures in the case of retropharyngeal abscess or removal of

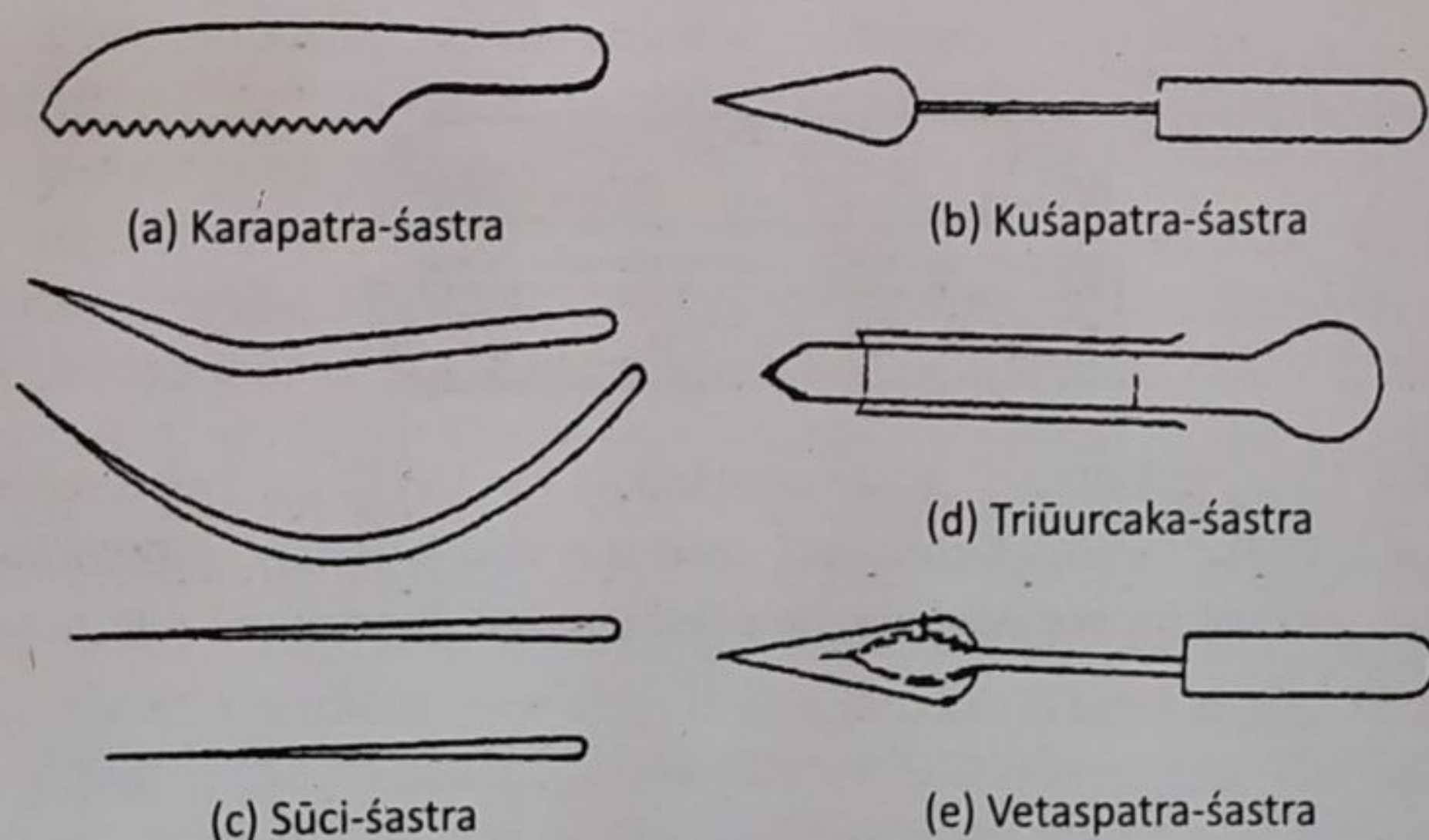


FIGURE 11.4 Some Sharp Instruments Mentioned in Suśruta-saṃhitā

Source: Sharma, A.N. (1997). "Surgical Techniques", Chapter 4 [4.4] in *History of Technology in India*, Vol. I, A.K. Bag (Ed.), Indian National Science Academy, New Delhi, p. 335. Reproduced with permission.

an opportunity for the ancient Indian surgeons to perform otoplasty (repair of the ear lobe) and rhinoplasty (surgery to change the shape of the nose). In *Suśruta-saṃhitā*, Chapter 16 of *Śarīrasthānam*, a procedure for repairing the nose that has been chopped off has been described in some detail¹⁹:

- ♦ The portion of the nose to be covered should be first measured with a leaf.
- ♦ A piece of skin flap of the required size should then be dissected from the cheek or the forehead and turned back to cover the nose.
- ♦ The flap of the skin taken from the cheek or forehead must be properly preserved to ensure proper blood circulation in the repaired area.
- ♦ The part of the nose is prepared well by making it raw.
- ♦ The surgeons should join the two parts quickly but evenly and calmly and keep the skin properly elevated by inserting two tubes inside the nostrils, so that the new nose may look comely.

11.6 SHIPBUILDING

Ancient Indians were very active in the maritime trade. Early Tamil literature of the Saṅgam period (before Common Era), foreign travellers' chronicles, and Roman authors have furnished details of Indian shipping and maritime trade. The Chola kings Raja Raja Chola I (985–1014 CE) and Rajendra Chola I (1014–1042 CE) carried maritime activities with the countries of the Far East and China. Studies suggest that Tamil Pandya embassies were received by Augustus Caesar and Roman historians mention a total of four embassies from the Tamil country.

In Tamil literature, we find references to different names of ships, classes of ships, and their parts, and other relevant information connected with navigation. Although there is no direct reference to the use of monsoon winds as an aid for sailing ships in early literature, the Buddhist Jātaka stories and Jain Canonical texts mention ships moving by force of the wind. Similarly, the Saṅgam period texts, viz. *Pura-nānūru*, *Aha-nānūru*, and *Madurai-kāñci* delineated different types of seagoing ships as they moved in the seas with the help of wind sails²⁰. However, we do not find much information on the art of shipbuilding.

- ♦ In Tamil literature we find references to different names of ships, classes of ships, and their parts, and other relevant information connected with navigation.
- ♦ The Jātaka stories as well as Buddhist accounts and paintings show that Buddhists were involved in maritime trade.

The Jātaka stories as well as Buddhist accounts and paintings show that Buddhists were involved in maritime trade. The caves of Ajanta, Aurangabad, and Ellora depict Bodhisattva Avalokiteśvara as a saviour of mariners in distressful conditions. The Pāli texts provide some information on Indian shipping. There are multiple references to the number of passengers that the ships carry, thereby giving us some idea of the size of the ships at that time. Prince Vijaya who was banished by his father sailed from Bengal in a fleet of ships carrying more than 700 passengers. Similarly, the ship in which the bride of Vijaya came to Ceylon is supposed to have carried nearly 800 persons. The ship of the Janaka-Jātaka carried 700 persons besides the Buddha and that of the Valāhassa-Jātaka carried 500 merchants. The only measurements given of a ship are those of the ship of the Śaṅkha-Jātaka which was 800 cubits (about 360 metres) in length, 600 (about 275 metres) in breadth, 20 fathoms (about 36.6 metres) in depth, and had three masts²¹.

TABLE 11.3 Different Types of Ships and Their Measurements given in Yukti-kalpataru

Ordinary (Sāmānya)				Special: Sea-going (Unnata)			
Name	Length	Breadth	Height	Name	Length	Breadth	Height
Kṣudrā (Diminutive)	4.57	1.14	1.14	Dīrghikā (Tall)	9.14	1.14	0.91
Madhyamā (Moderate)	6.86	3.43	3.43	Taraṇī (Moving hither and thither)	13.71	1.71	1.37
Bhīmā (Formidable)	11.43	5.71	5.71	Lolā (Moving hither and thither)	18.28	2.29	1.83
Capalā (Moving to and fro)	13.71	6.86	6.86	Gatvarā (Perishable)	22.85	2.86	2.29
Paṭalā (With covering)	18.28	9.14	9.14	Gāminī (Moving on)	27.42	3.43	2.74
Abhayā (Not dangerous)	20.57	10.28	10.28	Tarī (Running swiftly)	31.99	4.00	3.20
Dīrghā (Tall)	25.14	12.57	12.57	Jaṅghālā (Running Swiftly)	36.56	4.57	3.66
Patrapuṭā (Like a folded leaf)	27.42	13.71	13.71	Plāvinī (Flowing over)	41.13	5.14	4.11
Garbharā (With inner compartments)	31.99	16.00	16.00	Dhāriṇī (Power of possessing)	45.70	5.71	4.57
Mantharā (Curved)	34.28	17.14	17.14	Veginī (Having velocity)	50.27	6.28	5.03

* Source: Mukerji, R. (1977). "Notes on Ancient Hindu Shipping" in R.G. Bhandarkar Commemoration Volume, Bharatiya Publishing House, Delhi, pp. 447-455.

All values in metres. Conversion: 1 cubit = 0.457 metres

11.7 SIXTY-FOUR ART FORMS (64 KALĀS)

Kalā is defined as that which makes an individual happy and is an art form. The word 'art form' in the context of 'Kalā' has a wider meaning as it does not restrict itself only to performing arts. It includes a variety of specialised skills that are innate to human beings. By getting trained in these skills and by engaging in activities related to these, individuals can become happy and deliver a good job in whatever they do. This formed the bedrock of professional and trade classifications in earlier times. In the ancient Indian tradition, there is a concept of 64 kalās (known as catuḥ-ṣaṣṭi-kalā or vidyā) which captures the essence of various useful skills that individuals can develop. These could form the core of making a living for them. Table 11.4 summarises the 64 kalās organised under eight broad categories. As is evident from the table, this includes certain skills related to artistic pursuits, decoration, beautification of body, cooking and play, and entertainment.

What is however noteworthy is that the list of 64 kalās includes a wide range of engineering and technical skills. An examination of these skills provides further evidence of the use of S & T in ancient Indian society. Weaving and a host of other engineering skills including carpentry, metallurgy, and mineralogy suggest that there was enough application of these ideas in the society and therefore individuals who develop unique skills in these areas can make a living by professing these.

TABLE 11.4 A Summary of 64 Kalās

Type of Skills	No.
Artistic Skills: Singing, dancing, painting, theatrics, playing instruments, etc.	8
Skills in Cleaning and Beautifying Body: Dressing decorating, painting the face, applying aromatics, combing hair, setting ornaments, etc.	9
Skills of Decoration: Making a covering of flowers for a bed, making the groundwork of jewels, covering the bed, applying an admixture of colors, etc.	6
Cooking Related Skills	3
Play and Entertainment Skills: Playing on music in water, splashing with water, juggling, playing with thread, riddles, gambling, using children's toys, etc.	8
Skills for Intellectual Pursuits: Conversation, practicing language, composing, designing a literary work, lexicography and metres, reciting books, etc.	10
Engineering and Technical Skills: Needle works and weaving, spinning, carpentry, engineering, testing silver and jewels, metallurgy, mineralogy, practicing medicine, mechanics, etc.	14
Special Uncommon Skills: Knowing the mode of fighting of lambs, and birds, knowing conversation between male and female cockatoos, talking with fingers, prediction by heavenly voice, etc.	6

11.8 STATUS OF INDIGENOUS S & T

The available evidence and the scriptural references point to a culture that deployed S & T in various aspects of their day-to-day living. However, it is intriguing that we have no inkling of these today. We have very little awareness of these details. We also do not see the practice of any of these concepts in mainstream activities. Some of the S & T practices still in use have been branded as rural, tribal, and folk practices. They neither receive serious attention nor appreciation. A case in point is the current day water management practices. The trans-Yamuna – Sutlej canal is a multi-crore government initiative to bring water to the desert regions of Rajasthan. While this has not achieved its objectives, the age-old local practices to harvest water have been working well (see the video listed under the second bullet in “Discover IKS” exercise in Chapter 1 of the book for more details on this). In summary, we shall enumerate some broad observations concerning science and technology in ancient times and its current status:

- ◆ On the one hand, we find compelling evidence to conclude that ancient Indians had a good understanding of several aspects of science and technology, as we saw in Chapter 10. On the other hand, we are not able to get substantive details in the literature on various aspects of science, engineering and technology. The available information in some cases is broad. A case in point is the shipbuilding industry of ancient India. There are several

- ◆ There are several indications about the unique aspects of the Indian ships that travelled far and wide and conducted trade with several countries.
- ◆ Literary sources dealing with science, engineering and technology are not very detailed; substantive aspects of knowledge transmission continued in the oral tradition.

indications both in the Indian inscriptions and travellers' accounts of foreigners who visited India about the Indian ships and maritime travellers who travelled far and wide and conducted trade with several countries. This clearly indicates that they are adept in shipbuilding without which it may have been a dangerous activity to engage. There are intricate details and specifications involved in shipbuilding. However, we are unable to extract great details about shipbuilding from the Indian literature.

- ◆ As we know, Indian society relied substantially on an oral tradition for the preservation, and dissemination of knowledge across generations. The culture of extensive documentation and relying on written material as the source of knowledge is of fairly recent origin in India, dating back at most to half a millennium. Therefore, we find that literary sources dealing with science, engineering and technology are not very detailed and specific in nature. Perhaps substantive aspects of knowledge transmission continued in the oral tradition. The expertise of the teacher was often a closely guarded secret, available only to the students working with them on the job. Even today we find that there are some traditions in which knowledge is transferred from generation to generation (such as Astrologers, Vaidyas, and Śilpis).
- ◆ There was a well-developed caste system that was closely aligned to the skilled craft being practiced by the people. For example, all the metalsmiths and Goldsmiths were of a certain caste and so were the carpenters, the building mechanics, or the Alchemists, to name a few. These practices and know-how were maintained within the family and passed on from generation to generation.
- ◆ Large parts of India were subjected to foreign rulers for over 700 years since the beginning of 13th century CE. The priorities of what needs to be encouraged and in what manner were left entirely to the ruling class and this has also contributed substantially to the waning away of indigenous skills, practices, and know-how. For example, much of the knowledge about mining and metalworking that ancient Indians possessed was in grave danger when the British Raj imposed a mining ban in the 18th century. Furthermore, imposing production taxes also contributed to making metalworking practices economically infeasible. This enabled the British Raj to export a large chunk of ores for value addition in England.
- ◆ With the advent of modern governance and social systems introduced by the British Raj, many skills and practices went off the mainstream swiftly, thereby making these structures weaker and perhaps irrelevant. Thus, much of ancient art and craft perished over time for lack of institutional support and broad governmental and social acceptance. Moreover, it was not providing enough economic incentives to perpetuate this knowledge across generations. A few remaining skills in the country today suffer from huge survival challenges which are largely economic and institutional in nature.